Simulating Organized Group Behavior: New Framework, Benchmark, and Analysis

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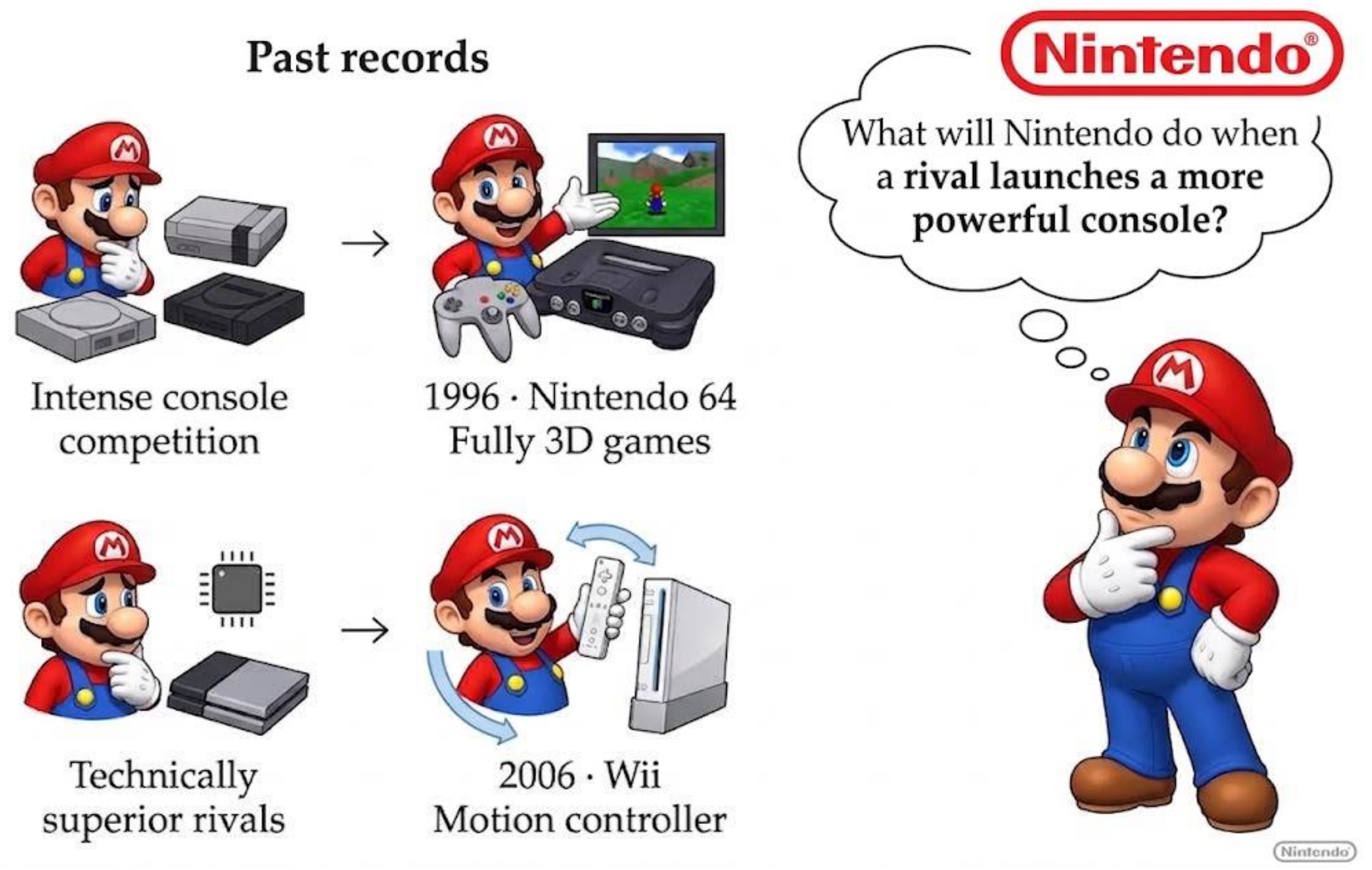
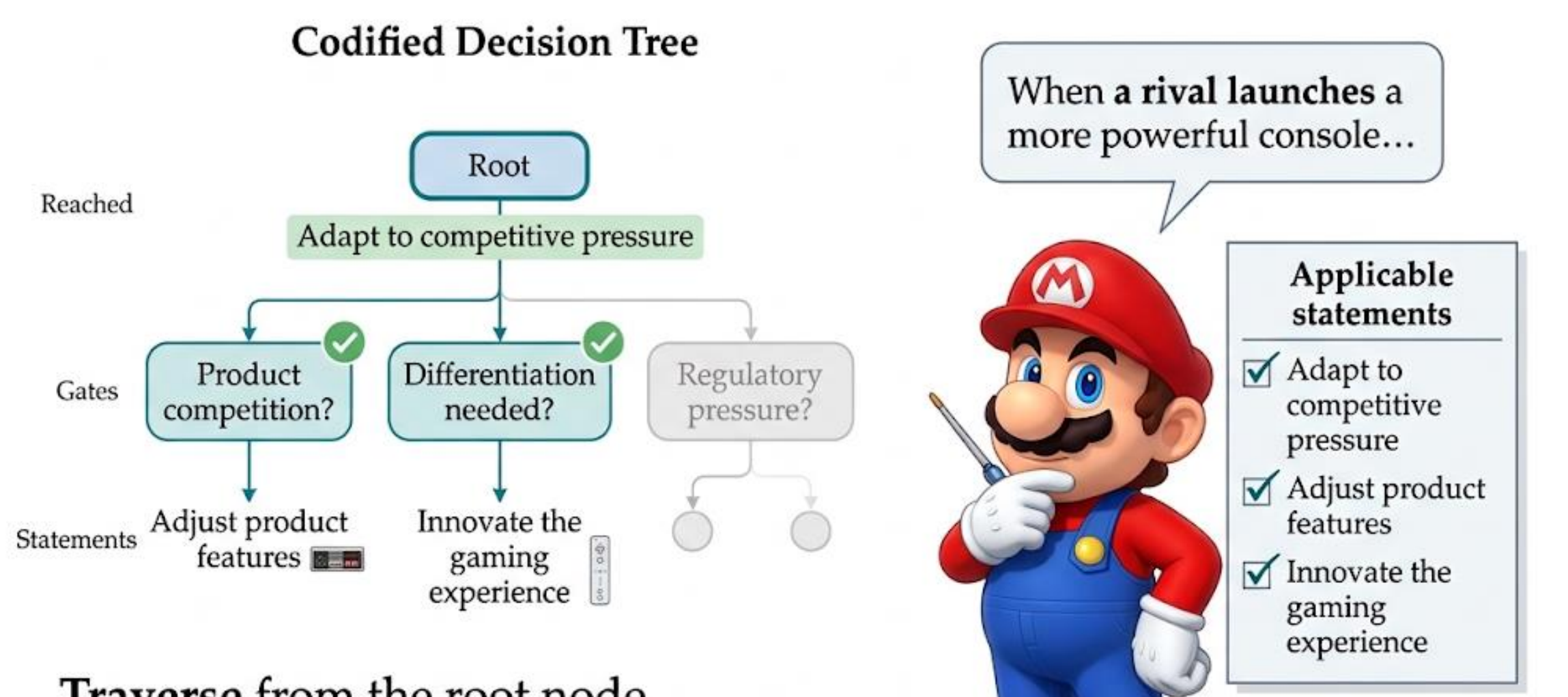
Project Page

Task: organized group behavior simulation


Past records
Intense console competition
1996 · Nintendo 64
Fully 3D games
Technically superior rivals
2006 · Wii
Motion controller
Nintendo
What will Nintendo do when a rival launches a more powerful console?

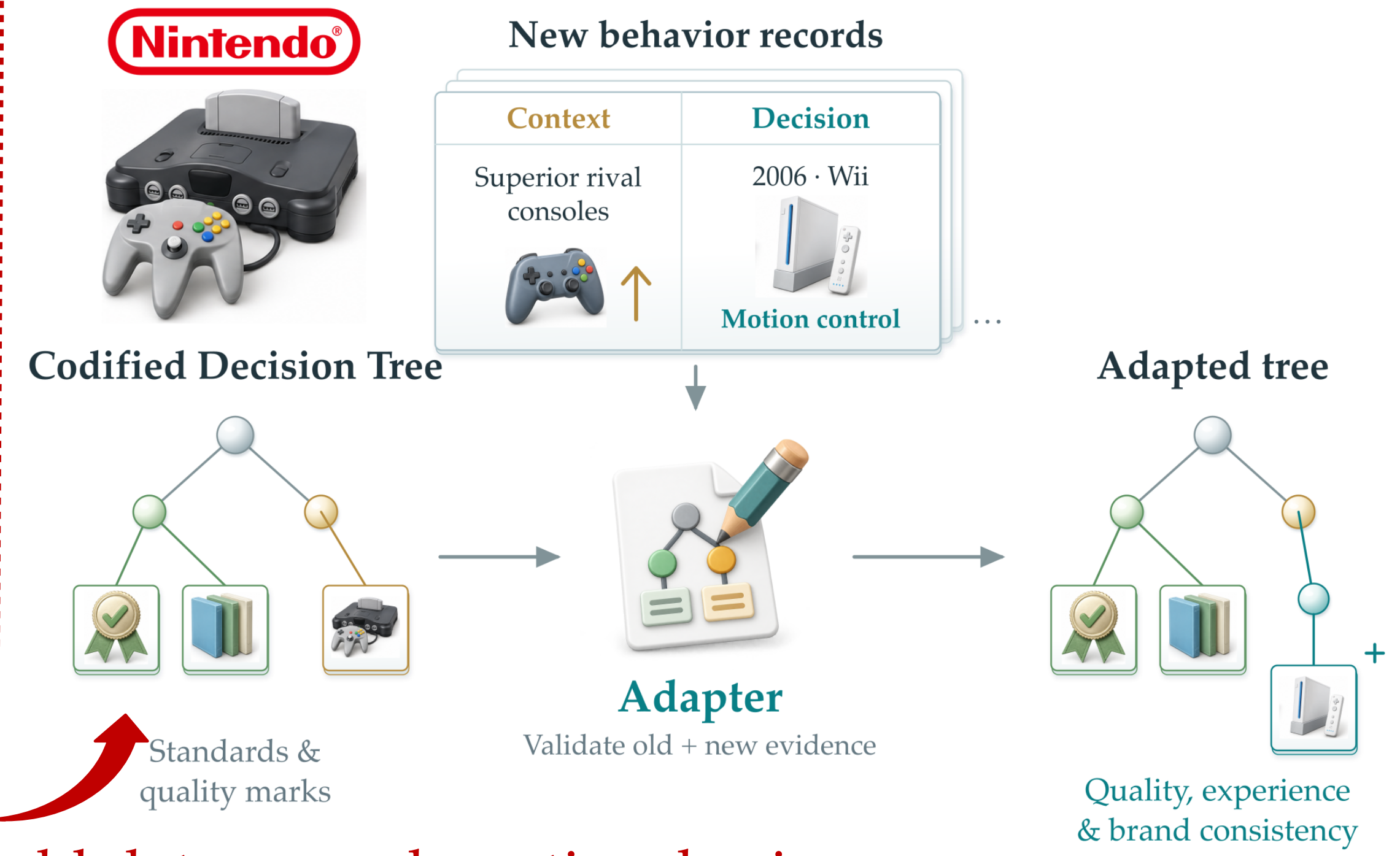
Codified Decision Tree

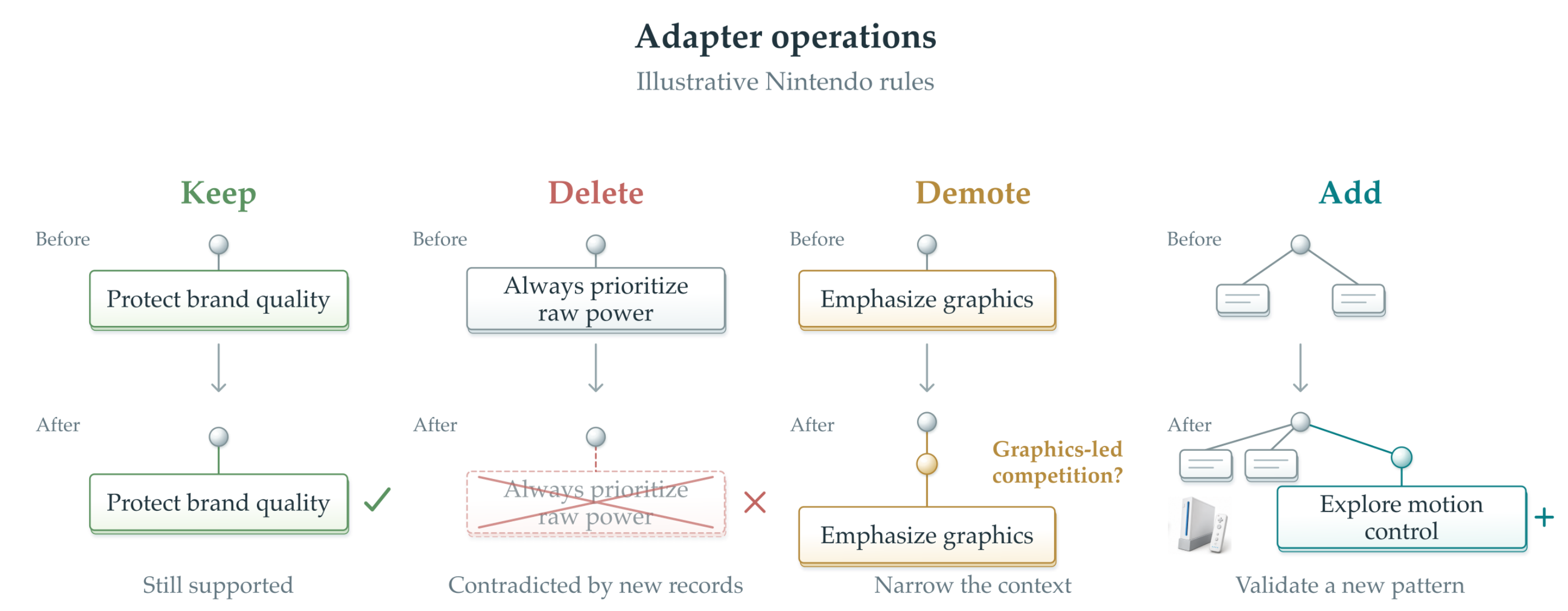

Codified Decision Tree
Reached
Root
Adapt to competitive pressure
Gates
Product competition?
Differentiation needed?
Regulatory pressure?
Statements
Adjust product features
Innovate the gaming experience
When a rival launches a more powerful console…
Applicable statements
☑ Adapt to competitive pressure
☑ Adjust product features
☑ Innovate the gaming experience

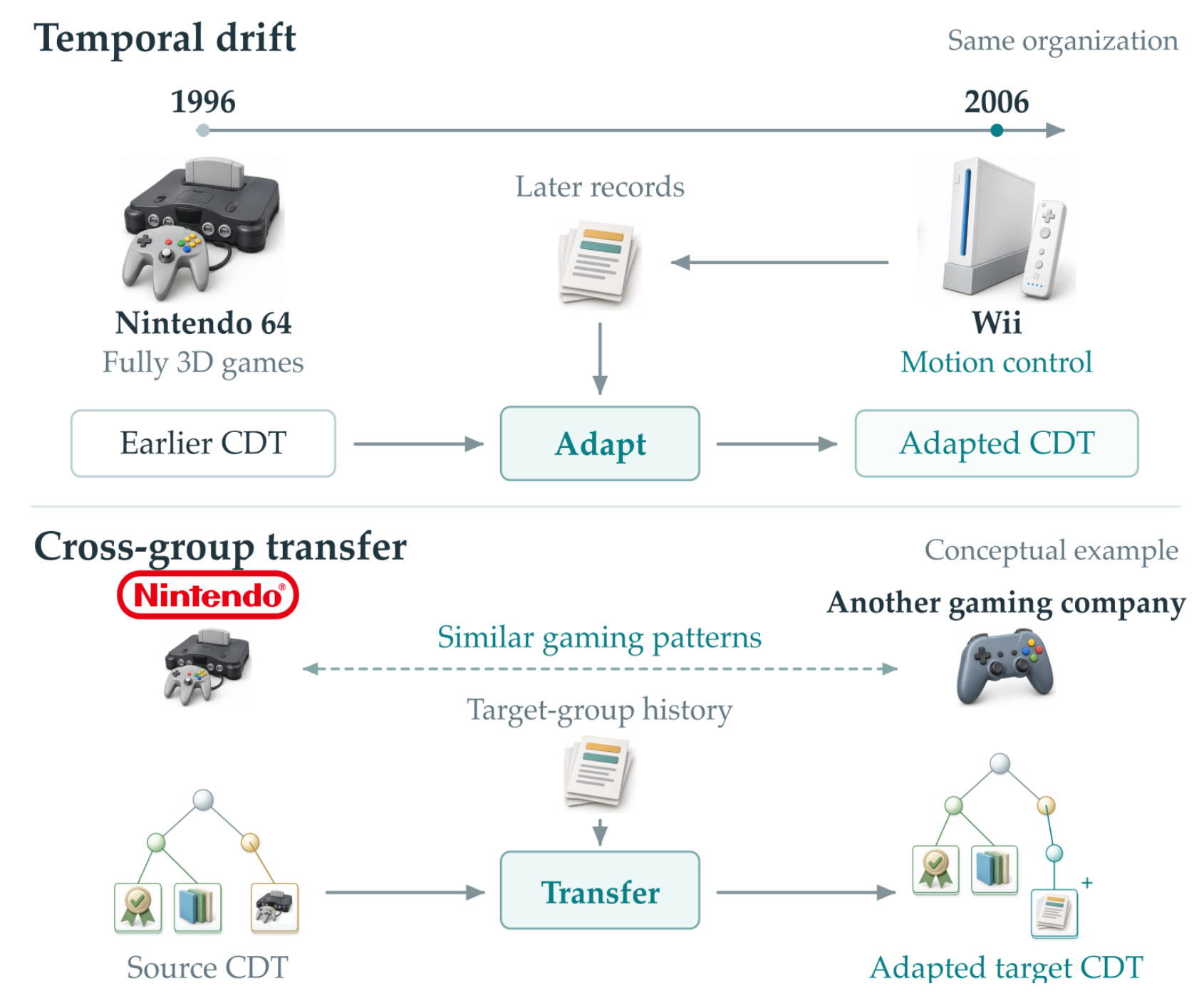
Traverse from the root node.
Check if the context aligns with the gate.
Collect applicable statements.

Q: Why Adapter? A: Real-world data spans long time horizons…

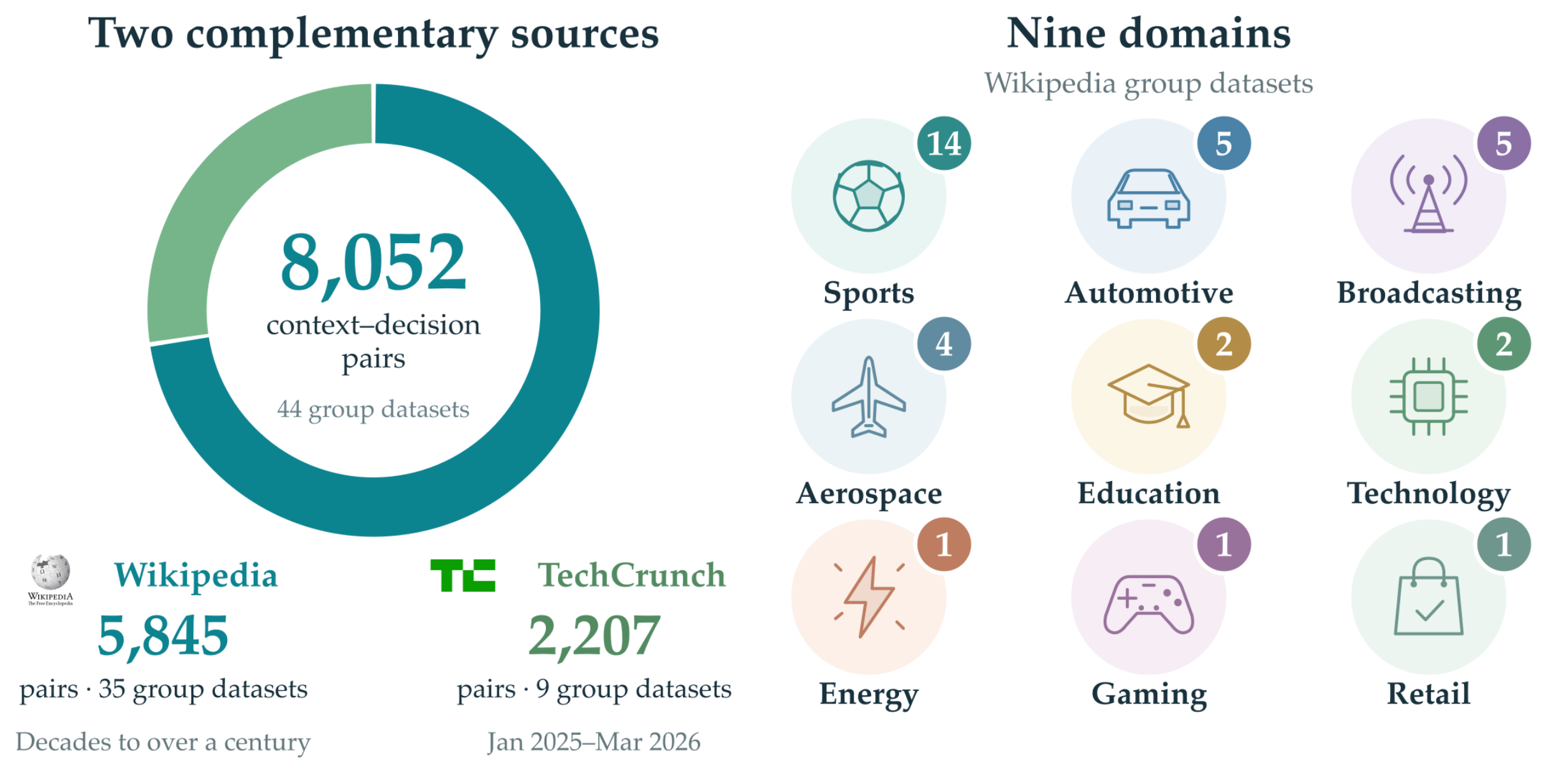
Adapter Mechanism



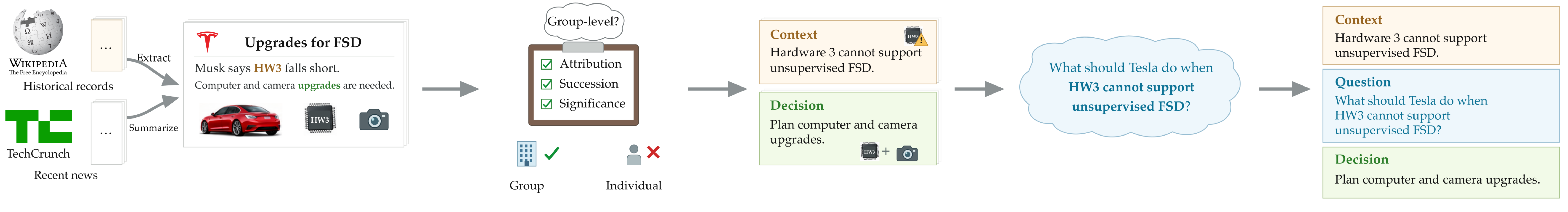




Benchmark Statistics



Benchmark Construction



Upgrades for FSD
Musk says HW3 falls short.
Computer and camera upgrades are needed.

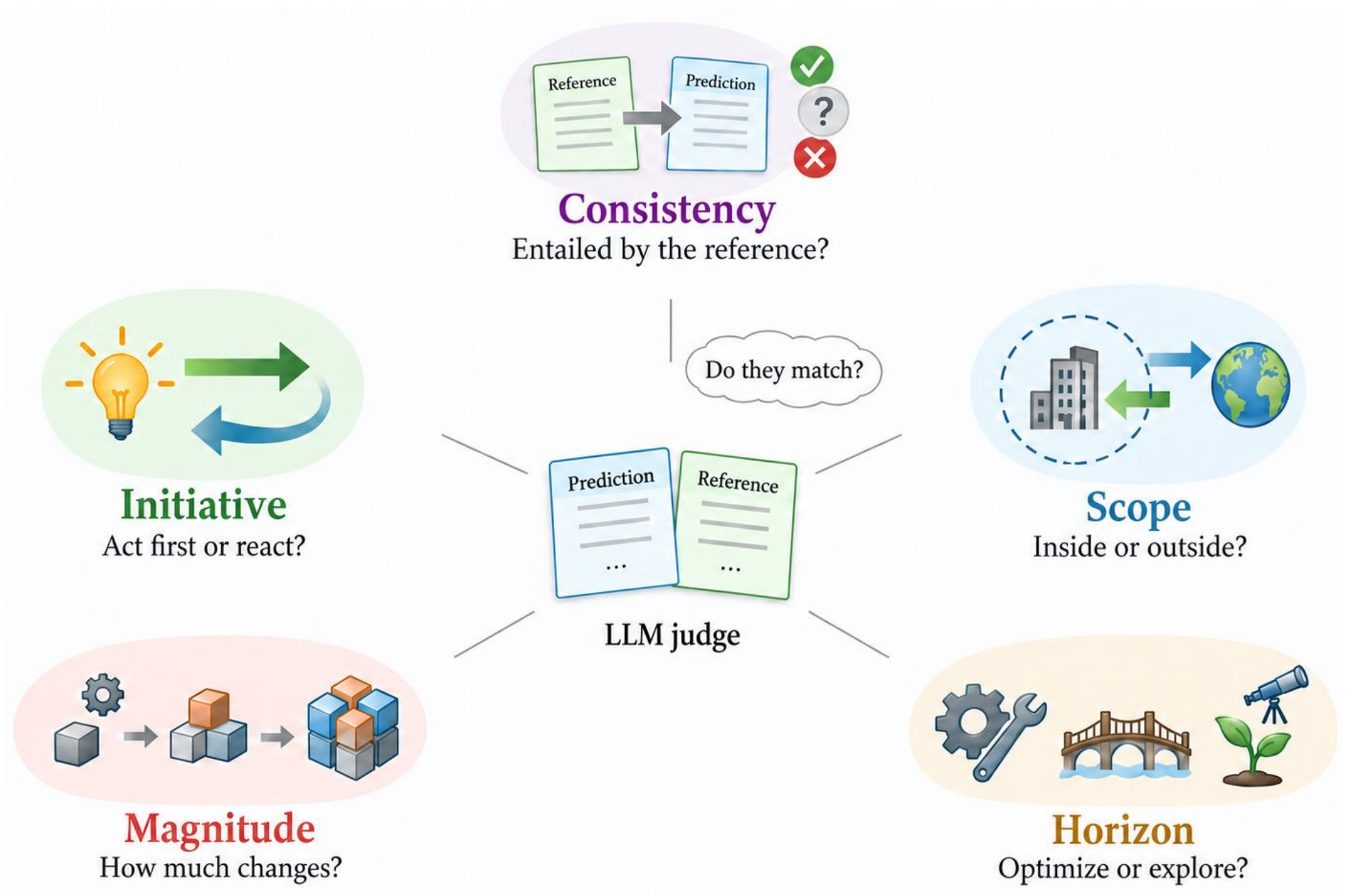
Context Hardware 3 cannot support unsupervised FSD.

Decision Plan computer and camera upgrades.

Question What should Tesla do when HW3 cannot support unsupervised FSD?

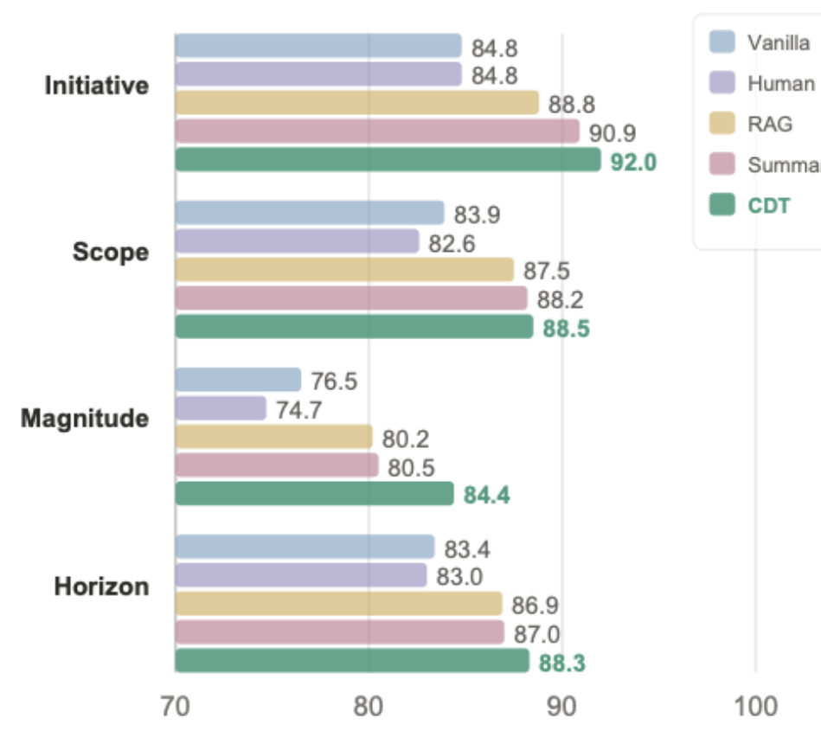
1 Extract / summarize — 2 Filter group actions — 3 Refine decisions — 4 Generate questions — 5 Example

Evaluation Metrics



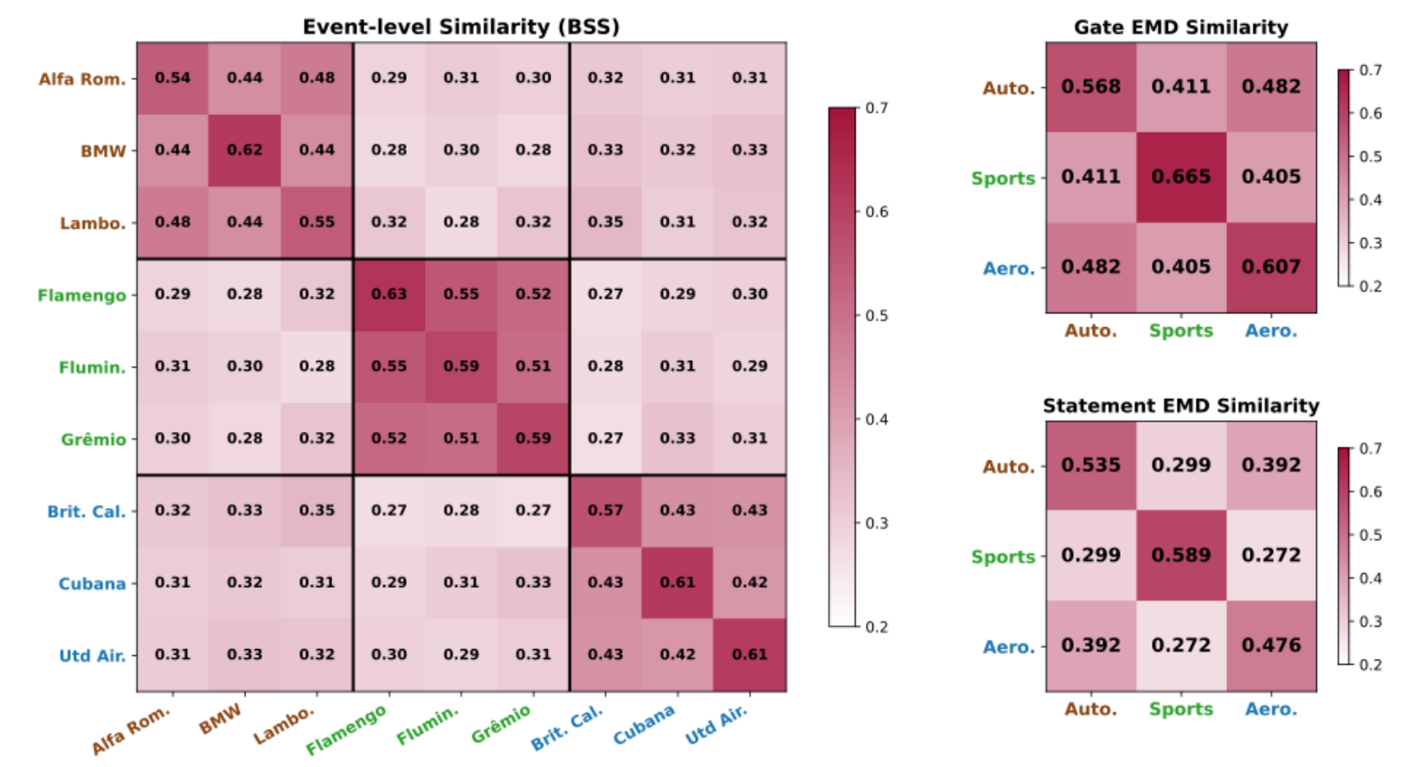
Experiments & Analysis

I. Structured representation (CDT) improves prediction



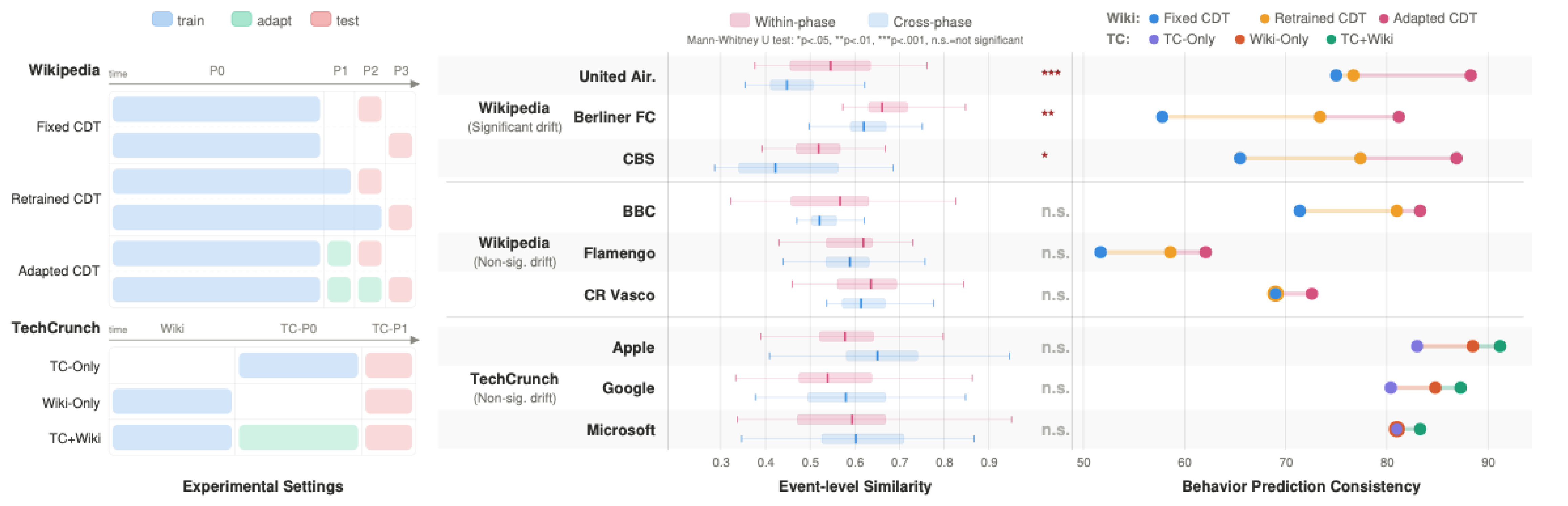
	Wikipedia										TC
Method	Aero.	Auto.	Broad.	Edu.	Ener.	Game	Ret.	Spo.	Tech.	*Avg*	*Avg*
Vanilla	67.7	62.9	64.4	69.4	52.1	59.2	61.1	54.4	62.4	61.5	78.5
LoRA Finetuning	54.8	52.8	52.2	56.2	57.3	50.0	48.6	43.4	59.2	52.7	67.6
Human Profile	**70.7**	65.4	63.1	68.1	55.2	62.5	55.6	53.9	63.9	62.0	79.8
Summarized-based	69.8	62.9	66.2	68.1	68.8	**65.0**	59.7	58.3	64.9	64.9	79.0
RAG	67.3	62.5	64.4	66.0	60.4	59.2	55.6	55.2	67.5	62.0	84.2
Ours (CDT)	70.0	**69.1**	**68.5**	**70.8**	**74.0**	**65.0**	**66.7**	**64.5**	**70.1**	**68.7**	**85.7**
Temporal Adaptation Study evaluation on P2 & P3											
Fixed CDT	70.2	65.0	69.0	67.7	75.0	62.5	**66.7**	65.2	**75.8**	68.6	–
Retrained CDT	71.2	67.5	73.4	66.7	73.4	62.5	54.2	66.7	67.2	67.0	–
Adapted CDT	**76.0**	**68.3**	**77.2**	**75.0**	**78.1**	**72.5**	64.6	**68.8**	68.8	**72.1**	–

III. Shared patterns enable behavior transfer



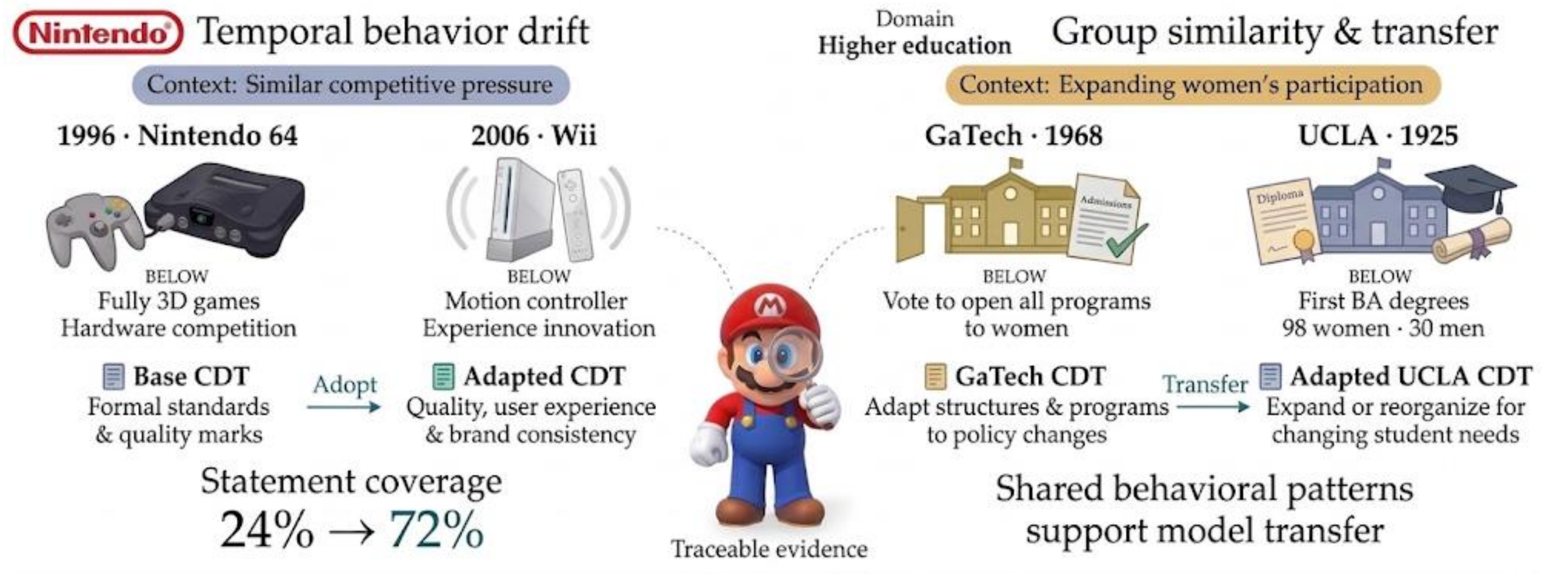
Domain	Source	Target	Vanilla	Tgt CDT	Transfer
Aero.	Brit. Cal.	Airbus	75.0	61.4	**83.3**
Auto.	Volkswagen	Ford	73.8	66.7	**76.2**
Spo.	Flumin.	Derry City	65.0	71.7	**80.0**
	SE_Palmeiras	GELP	66.7	64.6	**68.8**
Edu.	Cornell	Princeton	65.4	76.9	**84.6**
	Cornell	UPenn	53.3	60.0	**63.3**
	Cornell	Stanford	79.4	70.6	**82.4**
	GaTech	UCLA	62.5	62.5	**70.0**

II. Adaptation tracks behavioral drift and updates behavioral model



Case Studies

(left: temporal behavior drift; right: group pattern transfer)


Nintendo
Temporal behavior drift
Context: Similar competitive pressure
1996 · Nintendo 64
BELOW
Fully 3D games
Hardware competition
2006 · Wii
BELOW
Motion controller
Experience innovation
Base CDT
Formal standards & quality marks
Adopt
Adapted CDT
Quality, user experience & brand consistency
Statement coverage
24% → 72%
Traceable evidence
Domain
Higher education
Group similarity & transfer
Context: Expanding women's participation
GaTech · 1968
BELOW
Vote to open all programs to women
UCLA · 1925
BELOW
First BA degrees
98 women · 30 men
GaTech CDT
Adapt structures & programs to policy changes
Transfer
Adapted UCLA CDT
Expand or reorganize for changing student needs
Shared behavioral patterns support model transfer